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Roll no 20

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Date
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Q 1)

Q1) Similarities betⁿ Windows and Linux operating system

- 1) Multitasking : Both windows and linux support multitasking, allowing multiple applications to run simultaneously. This is crucial for enhancing user productivity.
For users this means smoother operation and better time management. For developers it allows better time management.
- 2) File system support : Both OS support various file systems. Windows uses NTFS, while Linux uses NTFS and FAT32.
Users benefit from efficient storage management.
- 3) Graphical User Interface (GUI) : Both offer a GUI making the OS accessible to users with varying levels of experience, enhancing its usability.
- 4) Networking Capabilities : Both support TCP/IP, DNS and DHCP which are essential for connecting to the internet and networks.
- 5) Security features : Both windows and linux provide built-in security features like firewalls, encryption and user authentication to protect against unauthorized access and data breaches.

Differences between

mono & hybrid

- 1) Kernel type : Windows uses monolithic kernel and Linux uses hybrid kernel. Kernel can be modified and optimized by users and developers. Linux provides more customization while windows a more uniform and stable environment.

- 2) **User interface Customization**: Linux allows extensive customization of the user interface, enabling users to change the desktop environment and system appearance. Windows offers limited options. Users who prefer personalization will prefer Linux.
- 3) **Software Compatibility**: Windows is more widely supported by commercial software, offering a broader range of applications, especially in gaming and professional use. Linux, however, excels in open source software and its development and ~~updates~~. Developers must consider customization while choosing Linux or Windows.
- 4) **Security model**: Linux is generally considered more secure due to its permission-based model and lower user base, which makes it a less attractive target. Windows is more targeted though it has extensive security measures. Windows developers must prioritize security in their software.
- 5) **Update management**: Windows manages updates centrally through Windows update. Linux, on the other hand, allows users to choose when and how to update their system.
Fact Linux users has more control over their Kernel or system.

Q2)

Function of the Bootloader :-

The boot loader is a critical piece of software that initializes when a computer starts up. Its primary function is to load the operating system kernel into memory and initiate the boot process.

Bootloader is responsible for:

- 1) **Initializing system Hardware:** Before loading the OS, the bootloader initializes the system hardware, ensuring that essential components like the CPU, memory and input/output systems are ready for operation.
- 2) **Loading the Kernel:** The boot loader locates the OS kernel on the storage device, loads it into memory and hands over control to the kernel, which then continues the boot process.
- 3) **Dual / Multiple boot management:** For systems with multiple OS installed, the boot loader provides a menu to select which OS to boot. This is useful for users who want to run different OS on the same machine.
- 4) **Passing parameters to the Kernel:** The boot loader can pass various parameters to the kernel, such as specifying the root filesystem or other kernel options that affect how the OS operates.
- 5) **Error handling and recovery:** In case of boot failure, the bootloader often provides recovery options or booting into safe mode.

Different Bootloaders

1) GRUB (Grand unified Bootloader)

Advantages

- i) **Flexibility:** GRUB supports a wide range of file systems and OS, making it highly versatile.

- ii) **Customization:** Users can easily configure GRUB to include custom boot parameters and scripts.

- iii) **User interface:** GRUB offers a graphical menu for selecting OS, which is user-friendly.

Disadvantages i) Complex configuration as its files can be complex and difficult for new users
ii) Size: GRUB is larger and more complex

2) LILO (Linux loader)

Advantages i) Simplicity: LILO is straight forward and simple to configure, making it accessible for users with basic needs

ii) speed: LILO is light weight and fast, ideal for systems where simplicity and speed are key

Disadvantages i) Limited features: LILO lacks many of the advanced features found in GRUB, such as support for modular filesystems and flexibility in boot options

ii) No graphical interface: LILO does not provide a graphical user interface, which can make it less user friendly for selecting bootloads

3) Windows Boot Manager:

i) Advantages: i) Integration: It is tightly integrated with Windows providing seamless booting into various Windows environments.

ii) Ease of use: The boot manager is automatically managed by Windows - requiring minimal user intervention

ii) Disadvantages i) Limited OS support: Windows Boot manager primarily supports booting into Windows OS

ii) Less Customization: Very less compared to GRUB
eNAND, USB

